



Rasterization



Rasterization is fast because triangles project to triangles: A 3D triangle always becomes a 2D triangle on screen. This geometric property makes the “which pixels are covered?” calculation extremely efficient.



Just as a farmer rakes soil into organized rows for planting, a Rasterizer rakes geometric shapes into organized pixels for display.

Rasterization turns scene entities into triangles.

Compared to Ray Tracing, it's a simpler method of rendering.

Triangles are the universal primitive in computer graphics because:

- They're always planar. Three points always define a plane
- They're simple to process mathematically
- Any complex surface can be approximated by many small triangles

Etymology

The word comes from two parts:

- "Raster" from Latin "rastrum" = "rake"
 - In Latin, a rastrum was a farming tool with teeth used to rake soil. This evolved in technology to mean:
 - In television technology: A raster is a rectangular grid of scan lines that an electron beam traces to create an image on a CRT screen
 - In computer graphics: A raster is a rectangular grid of pixels (like a bitmap)
 - Think of it as "raking" across the screen in horizontal lines
- "-ization" suffix meaning "process of making"
 - So Rasterization literally means "the process of converting something into a raster"
 - E.g. converting geometric data into a pixel grid